When does a spectral prior help graph learning? Connectivity-loss estimation under road-network disruptions

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Abstract

Rapid evaluation of many simultaneous road-link disruptions requires a useful compromise between exact spectral recomputation and local approximation. We estimate the relative loss of algebraic connectivity after multi-edge deletion using graph neural networks (GNNs) that learn a bounded correction to a first-order Fiedler sensitivity. The design is tested under independent, spatially clustered, and edge-betweenness-targeted failures, with graph-disjoint synthetic splits and zero-shot transfer to 13 OpenStreetMap (OSM) areas in six countries (48–1,259 nodes). GCN, GraphSAGE, and explicitly edge-aware MPNN backbones, feature ablations, and first- and truncated second-order analytical baselines isolate the contribution of the residual prior. OSM uncertainty uses an area-clustered hierarchical bootstrap with seed nested within area, rather than treating seeds or scenarios as geographic replicates. Across the expanded OSM panel, residual GCN improves spatial-failure MAE by 0.0391 (95% hierarchical interval 0.0151–0.0662), while residual GraphSAGE improves targeted-failure MAE by 0.0257 (0.0095–0.0446). Edge-MPNN residual differences are positive but imprecise. Second-order perturbation improves first-order MAE by only 0.0028–0.0053. Correction slopes fall from 0.84–0.95 under independent or spatial transfer to 0.51–0.58 under targeted transfer for GCN/GraphSAGE, directly quantifying residual shrinkage around a biased anchor. Controlled sparse-eigensolver scaling extends to 20,000 nodes and separates one-time spectral setup, amortized cost, and update-only cost. These results identify the spectral residual as a useful but domain-sensitive inductive bias, and delimit its applicability to structural connectivity rather than hazard or traffic-flow prediction. Code and cached networks are available at github.com/lee-vtruong/ResiliRoad.

Keywords: algebraic connectivity; Fiedler vector; graph neural network; spatial network; infrastructure robustness; domain shift.

1 Introduction

Road networks are spatially embedded complex networks whose link losses can fragment access routes and alter system-wide connectivity. Exhaustive exact evaluation becomes costly when a planner must screen many multi-link scenarios. Conversely, a first-order perturbation is fast and interpretable, but its local linearity can fail under finite deletions, eigenvalue crossings, and disconnection. This tension motivates a hybrid estimator: retain the known spectral direction and learn only its nonlinear error.

Algebraic connectivity, the second-smallest Laplacian eigenvalue, is a global structural quantity [1, 2]. It does not measure travel demand, congestion, capacity, or recovery. The narrower goal here is fast structural stress testing: given an intact weighted graph and a set of removed edges, estimate its relative algebraic-connectivity loss. This distinction is important because transport robustness has also been defined using system travel time, capacity loss, accessibility, and repair trajectories [6, 7, 13].

This study asks four questions. **RQ1**: Does learning a residual over a Fiedler approximation improve direct graph regression? **RQ2**: Does this advantage persist under correlated disruptions and geographic domain shift? **RQ3**: Which gains come from the explicit prior, Fiedler node feature, coordinates, or adjacency-aware message passing? **RQ4**: Where does the estimator fail and how does its runtime scale?

Rather than asking whether one new GNN architecture wins, the central thesis is *when a mathematically informed spectral prior helps graph learning under distribution shift, and when it becomes a biased anchor*. The contributions are: (i) a bounded residual spectral construction for simultaneous edge deletions; (ii) controlled feature, GCN/GraphSAGE/edge-MPNN, second-order, and reliability-context ablations; (iii) three disruption processes on 13 cached OSM networks in six countries; (iv) both synthetic-to-real zero-shot and geographically disjoint OSM transfer; (v) area-clustered hierarchical inference, calibration, eigengap and correction diagnostics; and (vi) a benchmark separating eigensolver setup, amortized spectral evaluation, and update-only cost. Importantly, the transfer experiments reveal a limitation: the residual prior is not uniformly superior once OSM training data are available.

2 Related work

2.1 Spectral robustness of complex networks

Fiedler introduced algebraic connectivity as a graph invariant [1]; later work related Laplacian spectra to expansion, connectivity, synchronization, and robustness [2, 3, 5]. Capacity-weighted spectral analysis has also been used to diagnose transport-network vulnerability [16]. For a simple eigenvalue, the squared difference of Fiedler coordinates across an edge gives the derivative with respect to its weight [4]. Such derivatives explain infinitesimal sensitivity, whereas the present task includes several finite edge deletions. The residual model specifically targets the missing interaction and higher-order terms rather than replacing spectral analysis.

2.2 Road-network disruption

Road robustness depends on graph representation and disruption process. Multi-granularity analyses of empirical city networks show that structural conclusions can change with representation [8]. Link-based indices using flow or capacity address functional effects [6, 7], while critical-scenario optimization searches for damaging link combinations [9]. Hazard-independent studies compare random, localized, and targeted multi-link failures [12]; temporal analysis of Zurich also demonstrates process-dependent robustness [11]. Random road graph models enable controlled topology experiments [10]. Our spatial-cluster mechanism belongs to this stress-testing tradition but is a geometric proxy, not an empirical hazard model.

Recent evidence makes scale and task definition especially important. Jana et al. [18] use an edge-based GNN to rank critical road segments after disruption, directly motivating learning-based rapid screening; their output is an edge ranking, whereas ours is a graph-level spectral-loss estimate. Boeing and Ha [19] simulate 2.4 billion trips across more than 8,000 urban areas in 178 countries, establishing an empirical scale far beyond our 13 neighbourhood networks. Zang et al. [20] predict dynamic traffic resilience under rainfall with a multi-granularity GNN; their functional, temporal target is complementary to our static structural target. Reviews distinguish connectivity, robustness, vulnerability, and recovery-based resilience and warn against treating them as synonyms [17, 15, 21].

2.3 Graph learning and analytical priors

GCNs aggregate local neighbourhood information and are permutation equivariant at node level [22]. GraphSAGE supplies an inductive aggregation backbone [23]; GIN studies the expressive limits of neighbourhood aggregation [26], while attention and message-passing frameworks expose alternative node/edge interactions [25, 24]. Deep Sets provides an invariant alternative that does not propagate along edges [27]. Graph Networks have also been used as learnable physical simulators [28], illustrating how known structure and learned corrections can coexist. The novelty claimed here is not the first use of GNNs, residual learning, or spectral metrics. It is the leakage-resistant empirical separation of an explicit spectral prior from a spectral node feature, followed by geographic transfer and negative-ablation analysis on spatial infrastructure networks. Edge-aware modelling is especially relevant because disruption acts on links: Jana et al. [18] encode a road line graph for edge ranking, and Almeida et al. [29] report edge-aware attention for backbone-network load prediction. The present study instead asks whether a simple node-message-passing estimator benefits from an explicit spectral prior across domains. Recent JCN work on urban edge removal uses diffusive transport [30], situating algebraic connectivity among several legitimate network-function proxies.

3 Problem formulation and method

Let $G = (V, E, w)$ be a connected undirected weighted graph, with combinatorial Laplacian $L = D - A$ and eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots$. For disrupted edges $S \subseteq E$, the target is

$$y(G, S) = \mathrm{clip}\left(1 - \frac{\lambda_2(G - S)}{\lambda_2(G)}, 0, 1\right). \tag{1}$$

Thus a disconnected damaged graph has $y = 1$. If u_2 is a unit Fiedler vector and λ_2 is simple, edge sensitivity is

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (u_{2,i} - u_{2,j})^2. \tag{2}$$

The analytical prediction sums intact-graph derivatives:

$$\widehat{y}_{\mathrm{spec}} = \mathrm{clip}\left(\frac{\sum_{(i,j)\in S} w_{ij}(u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1\right). \tag{3}$$

Each damaged graph supplies normalized adjacency and node features: intact degree, failed-edge incidence, absolute normalized Fiedler coordinate, and two normalized coordinates. Three 48-unit GCN layers feed mean–max graph pooling and a two-layer head. The direct model predicts y; the residual model predicts

$$\widehat{y}_{\text{res}} = \text{clip}\left(\widehat{y}_{\text{spec}} + \tanh r_\theta(G, S), 0, 1\right). \tag{4}$$

The bounded correction concentrates capacity on finite-deletion error but can also inherit domain-specific bias from the prior.

To answer whether learning is needed beyond a higher-order analytical model, a truncated perturbation baseline retains the 12 lowest Laplacian modes:

$$\Delta\lambda_2^{(2)} = u_2^\top \Delta L u_2 + \sum_{k \neq 2} \frac{|u_k^\top \Delta L u_2|^2}{\lambda_2 - \lambda_k}. \tag{5}$$

We also record the relative eigengap $\gamma = (\lambda_3 - \lambda_2)/\lambda_2$, which measures proximity to the next mode but is not supplied to any learner.

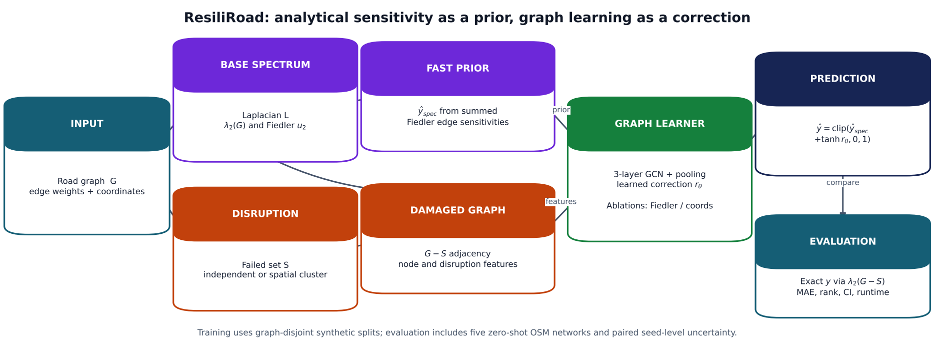

Figure 1: ResiliRoad workflow. Exact damaged-graph eigendecomposition constructs labels only. The intact Fiedler vector produces both an optional node feature and the explicit first-order prior; the GCN learns a bounded correction.
Alt text: A flow diagram splits an intact road graph into an exact-label branch and a fast spectral-prior branch. Damaged adjacency and node features enter a GCN whose residual is added to the spectral estimate.

The ablation matrix contains direct and residual GCNs with all features, without Fiedler, and without coordinates; Deep Sets and a summary-statistic MLP are non-GCN baselines. A further reliability-context residual appends the prior, normalized failure count, density, and graph size to the prediction head. It does not use damaged connectivity or any label-derived flag. As a stronger architecture control, GraphSAGE concatenates each node state with a normalized neighbourhood aggregate before each of three learned transformations. Its direct and residual variants use the same features, pooling, optimizer, split, and stopping rule as the GCN. The edge-aware

MPNN instead forms $m_{ij} = \phi(h_i, h_j, e_{ij})$ over every intact candidate link, where e_{ij} contains normalized conductance, inverse-conductance length, a failed edge indicator, and normalized $(u_{2,i} - u_{2,j})^2$. Mean incoming messages update node states in three layers. Keeping failed links as candidates allows the model to observe deletion explicitly rather than only through damaged adjacency and node incidence.

4 Experimental design

4.1 Synthetic graphs and disruption processes

Connected random geometric graphs contain 35–65 nodes and inverse-length edge conductance. The original two-mode experiment generates 800 scenarios per mode and seed; the expanded three-mode experiment generates 400 per mode and seed. Entire base-graph families are assigned to train, validation, or test. Each scenario removes one to eight edges. Independent failure samples edges uniformly. Spatial-cluster failure chooses a random epicentre and removes the required number of edges with nearest geometric midpoints. Both modes share the same failure-count distribution. The third regime samples without replacement with probability proportional to unweighted edge-betweenness centrality. It targets globally important links without using the Fiedler sensitivity that defines the analytical prior.

4.2 OpenStreetMap networks

OSMnx [32] was used to obtain drivable networks around 13 areas in Vietnam, Singapore, Malaysia, Thailand, Taiwan, and Japan. Five 650-m neighbourhoods preserve the original evaluation; eight 1.2–1.6-km areas add larger and morphologically diverse networks. Directed multigraphs were simplified, converted to undirected simple graphs, and restricted to the largest connected component. Conductance is $100/\text{length}_{\text{m}}$. Cached GraphML freezes the exact instances because live OSM data evolve.

Table 1: OpenStreetMap networks used in the study.

Area	Country	Radius (m)	Nodes	Edges	Density
HCMUS	Vietnam	650	163	221	.0167
VIASM	Vietnam	650	206	301	.0143
Da Nang	Vietnam	650	142	203	.0203
Can Tho	Vietnam	650	114	178	.0276
Da Lat	Vietnam	650	48	55	.0488
Singapore	Singapore	1,400	759	1,121	.00390
Kuala Lumpur	Malaysia	1,500	730	1,059	.00398
George Town	Malaysia	1,600	936	1,334	.00305
Bangkok	Thailand	1,400	1,072	1,457	.00254
Chiang Mai	Thailand	1,600	1,259	1,661	.00210
Taipei	Taiwan	1,300	965	1,546	.00332
Kyoto	Japan	1,500	909	1,468	.00356
Tokyo	Japan	1,200	1,108	1,675	.00273

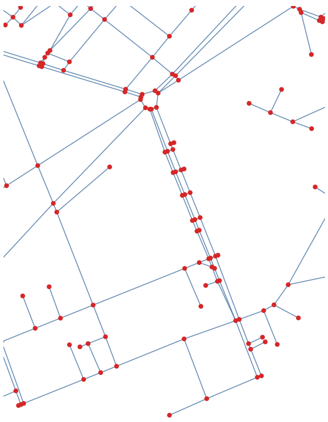

Figure 2: Processed HCMUS base network (163 nodes, 221 edges). Disruption scenarios remove subsets of the displayed links. Map data copyright OpenStreetMap contributors.

Alt text: A sparse, irregular street graph is drawn with red intersection nodes and pale-blue road edges; a dense central corridor connects several branching neighbourhoods.

Two transfer protocols answer different questions. In *zero-shot OSM*, all learned models train only on synthetic graphs and test on 100 scenarios per site, mode, and seed. In *leave-one-area-out OSM*, one site is test, the next site in a fixed rotation is validation, and the remaining three sites are training. Forty scenarios per site and mode are generated for each of five seeds; the direct and residual GCNs train for at most 25 epochs. No base area is shared across train, validation, and test in a fold. An expanded zero-shot protocol trains GCN, GraphSAGE, and edge-MPNN direct and residual variants on 400 synthetic scenarios per failure mode and evaluates 20 scenarios per area, mode, and seed across all 13 OSM areas. The original and expanded protocols are reported separately; the latter supplies geographic rather than merely random-seed replication.

4.3 Statistics, calibration, and runtime

For synthetic graph populations, seeds 11, 22, 33, 44, and 55 define independent generator replicates. For OSM, the road area is the outer sampling cluster and seed is nested within area. Hierarchical intervals resample areas with replacement and then one seed realization within each sampled area 10,000 times. Paired comparisons preserve method pairing within area and seed. The five-area leave-one-area-out analysis is re-estimated with the same area-outer hierarchy. No scenario-level p-value is used. Continuous calibration is summarized by regressing target on prediction within each seed, reporting slope, intercept, and signed mean error. Error slices by connectivity, failure count, density/site, and spectral clipping are diagnostic.

The original OSM benchmark uses dense symmetric eigendecomposition and one-scenario neural inference on CPU. A separate controlled benchmark uses sparse symmetric eigensolves on random geometric graphs with 50–800 nodes. It reports exact damaged-graph solution, update-only Fiedler sensitivity, and amortized spectral cost (one intact setup spread over ten scenarios). These denser graphs are computational stress tests rather than road-realistic samples.

5 Results

5.1 Graph-disjoint synthetic and zero-shot OSM results

Table 2: MAE as seed mean [95% seed-bootstrap interval].

	Synthetic		Zero-shot OSM	
Method	Independent	Spatial	Independent	Spatial
Spectral	.069 [.048,.095]	.144 [.102,.183]	.502 [.482,.521]	.650 [.643,.657]
Summary MLP	.065 [.051,.079]	.108 [.082,.135]	.281 [.240,.320]	.347 [.269,.425]
Deep Sets	.068 [.058,.078]	.105 [.085,.126]	.114 [.105,.123]	.143 [.122,.164]
Direct GCN	.077 [.055,.098]	.081 [.067,.097]	.106 [.100,.112]	.102 [.077,.148]
Residual GCN	**.052 [.036,.070]**	**.065 [.049,.079]**	**.097 [.091,.102]**	**.090 [.071,.124]**

The residual model improves direct-GCN MAE by paired differences 0.0253 [0.0106, 0.0400] and 0.0160 [0.0048, 0.0241] on synthetic independent and spatial failures. The improvements remain positive in zero-shot OSM: 0.0085 [0.0039, 0.0131] and 0.0121 [0.0026, 0.0247]. Yet the analytical prior alone has poor OSM calibration despite useful rank information. Removing Fiedler or coordinates from residual-full yields intervals containing zero; individual node-feature necessity is therefore not established.

5.2 Geographically blocked transfer

Table 3: Leave-one-area-out OSM performance, area mean [95% hierarchical area-outer interval]. Each fold uses three training, one validation, and one test area.

Method	Independent MAE	Spatial-cluster MAE
Direct GCN	**.135 [.089,.199]**	**.109 [.067,.158]**
Residual GCN	.161 [.089,.319]	.144 [.059,.352]

The point-estimate ranking reverses when training is moved into the OSM domain. Direct GCN has mean R^2 0.714 and 0.747, compared with 0.550 and 0.529 for the residual model. The paired residual-minus-direct MAE differences are 0.0261 [-0.0357, 0.1572] and 0.0350 [-0.0479, 0.2320]; both area-outer intervals include zero, so five areas do not establish direct-model superiority. Residual performance is seed-sensitive: independent-failure MAE ranges from 0.092 to 0.265. Calibration explains part of this instability. Direct-GCN slopes stay between 0.91 and 1.06; residual slopes fall to 0.58 and 0.69 in the two poorest seeds, with substantial underprediction. Thus the spectral prior is helpful under one training distribution but can become a shortcut or a biased anchor under another.

5.3 Reliability-context negative ablation

Adding graph size, density, failure count, and spectral estimate to the residual head reduces synthetic spatial MAE by 0.0193 [0.0120, 0.0266] and has an inconclusive -0.0034 [-0.0159, 0.0111] change for synthetic independent failure (negative means improvement). It increases zero-shot OSM spatial MAE by 0.0304 [0.0067, 0.0579]; the independent increase is 0.0452 [-0.0029, 0.1019]. Simple reliability metadata therefore encourages distribution-specific calibration rather than robust transfer.

5.4 Backbone robustness

Replacing GCN propagation with GraphSAGE tests whether the main conclusion is an artefact of a weak backbone. Table 4 shows that this is not the case: residual learning improves point-estimate MAE for both backbones in all four zero-shot conditions. Paired direct-minus-residual differences for GraphSAGE are 0.0197 [0.0071, 0.0332] and 0.0190 [0.0107, 0.0273] on synthetic independent and spatial failures, and 0.0069 [0.0032, 0.0096] and 0.0239 [-0.0073, 0.0697] on OSM. The final OSM-spatial interval crosses zero, so the effect is not declared established there.

Table 4: Architecture control: MAE, seed mean [95% seed-bootstrap interval].

Domain	Failure	Direct GCN	Residual GCN	Direct GraphSAGE	Residual GraphSAGE
Synthetic	Independent	.077 [.054,.096]	.052 [.036,.071]	.066 [.050,.081]	**.046 [.034,.063]**
Synthetic	Spatial	.081 [.067,.097]	**.065 [.049,.079]**	.084 [.068,.101]	.065 [.047,.085]
OSM	Independent	.106 [.100,.111]	**.097 [.091,.102]**	.104 [.098,.108]	.097 [.091,.103]
OSM	Spatial	.102 [.077,.148]	.090 [.071,.124]	.113 [.082,.164]	**.089 [.079,.099]**

5.5 Cross-country, targeted, and edge-aware evaluation

Table 5 reports the expanded 13-area experiment. Positive paired differences favour the residual formulation. Residual point estimates improve all nine backbone–failure combinations, but area-level uncertainty is material: intervals exclude zero only for GCN under spatial failure and GraphSAGE under independent and targeted failure. The edge-aware MPNN does not outperform the node-message-passing models; for example, its residual MAE is .163, .199, and .156, compared with residual-GCN MAE .085, .057, and .135. Thus explicit failed-edge attributes are an architecture control, not an automatic accuracy gain.

Table 5: Expanded zero-shot OSM paired improvement (direct minus residual MAE), area mean [95% hierarchical interval].

Backbone	Independent	Spatial cluster	Targeted betweenness
GCN	.0167 [-.0019,.0338]	**.0391 [.0151,.0662]**	.0097 [-.0095,.0268]
GraphSAGE	**.0253 [.0114,.0425]**	.0159 [-.0043,.0365]	**.0257 [.0095,.0446]**
Edge-MPNN	.0042 [-.1210,.1227]	.0128 [-.1489,.1575]	.0359 [-.0412,.1113]

5.6 Eigengap, analytical order, and learned correction

The truncated second-order perturbation baseline improves over first order by .00284 [.00050,.00673], .00359 [.00081,.00756], and .00533 [.00133,.01187] MAE under independent, spatial, and targeted disruption, respectively. These gains are consistent but too small to close the gap to the best learned estimators, answering why a learned correction remains useful.

The relative eigengap $\gamma = (\lambda_3 - \lambda_2)/\lambda_2$ is informative but not a sufficient gate. First-order OSM MAE is .444, .369, and .422 in low, middle, and high graph-level eigengap tertiles: near-degeneracy is harder than the middle tertile, but the relationship is non-monotone. Residual improvement also does not vanish uniformly at small gaps.

Mechanistic correction diagnostics support a more specific biased-anchor interpretation. Under targeted transfer, regression slopes of learned on needed correction are .512 (GCN), .580 (GraphSAGE), and .486 (edge-MPNN), compared with .843, .834, and .617 under independent failure. The residual therefore tends to shrink the correction precisely in the targeted regime, rather than fully undoing a miscalibrated analytical anchor.

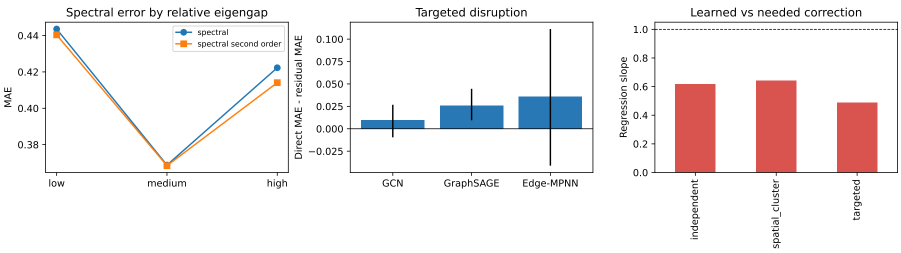

Figure 3: Expanded diagnostic results. (a) Area-level direct-minus-residual MAE with hierarchical intervals. (b) first- and second-order spectral error across relative-eigengap tertiles. (c) learned versus needed residual correction.

Alt text: Three panels compare residual improvements across backbones and disruptions, spectral errors across eigengap groups, and the calibration of learned corrections against ideal corrections.

5.7 Runtime and error regimes

On the five small OSM networks, exact dense eigendecomposition averages 5.094 ms per scenario, first-order update 0.018 ms, and residual inference 0.674 ms on CPU. Controlled sparse scaling gives exact times 8.37, 13.41, 14.62, 37.03, and 92.61 ms for 50, 100, 200, 400, and 800 nodes. Update-only time remains 0.088–0.113 ms; amortized spectral cost is 0.94, 0.89, 1.68, 3.51, and 8.79 ms, while residual-GCN inference is 0.67, 0.70, 0.96, 2.03, and 5.92 ms. The slight non-monotonicity at small size reflects fixed solver overhead and graph-to-graph variability; this benchmark is descriptive, not a complexity proof.

To test the actual screening motivation, a second sparse benchmark uses connected planar road-like graphs with 1,000, 2,000, 5,000, 10,000, and 20,000 nodes, with edge-to-node ratio $|E|/|V| \approx 2.1$, corresponding to mean degree $\bar{d} \approx 4.2$. Nodes occupy a normalized, near-square regular lattice. Horizontal and vertical nearest-neighbour edges form a connected truncated grid; each eligible cell independently receives its northwest diagonal with probability 0.12, which preserves planarity. Edge conductance is inverse Euclidean length. Each timed disruption removes eight uniformly sampled edges; base-graph connectivity is guaranteed by the grid backbone. At 20,000 nodes, sparse exact recomputation requires 1,302.4 ms per damaged graph. Intact eigensolver setup costs 1,335.3 ms once; amortized over 1,000 screened scenarios, the prior costs 1.41 ms, sparse GNN inference 21.76 ms, and the combined path 23.17 ms. Thus the measured screening path is 56 times faster at the largest size, conditional on reusing one intact graph. Only three scenarios on each of two generated graphs are timed per size, and the large graphs are not used for accuracy training; these measurements support computational scaling, not predictive generalization.

In zero-shot OSM, residual-full MAE is 0.070 for connected damaged graphs and 0.112 after disconnection. Site errors range from 0.041 (Can Tho) to 0.189 (Da Lat). The densest site is also the smallest and hardest, so density cannot be interpreted causally. No evaluated first-order estimate clips at one; the requested clipped/unclipped comparison is therefore undefined. The largest leave-one-area-out errors are retained in a machine-readable table instead of being removed as outliers.

6 Discussion

The central result is a condition, not a new-architecture claim. A spectral residual improves point estimates for GCN, GraphSAGE, and edge-MPNN on the expanded zero-shot OSM test, although only three of nine area-level

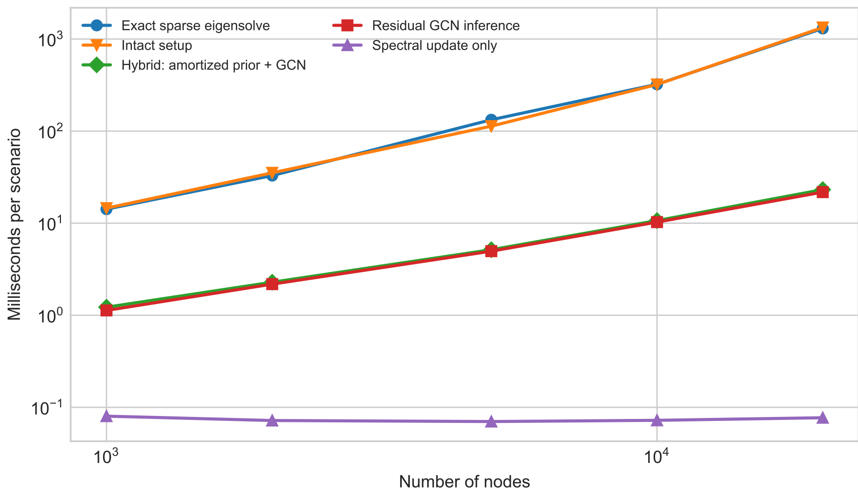

Figure 4: Sparse CPU scaling from 1,000 to 20,000 nodes. Exact recomputation is timed per damaged graph; the hybrid curve combines intact spectral setup amortized over 1,000 scenarios with first-order evaluation and sparse GNN inference. Points are means over two graphs and three scenarios per graph.

Alt text: A log–log runtime plot shows exact sparse eigendecomposition rising to about 1.3 seconds at 20,000 nodes, while the amortized hybrid path rises to about 23 milliseconds.

intervals exclude zero. It combines the prior's ranking signal with learned nonlinear calibration. However, blocked OSM transfer and reliability-context ablation show that the same inductive bias can amplify domain-specific error. The prior then behaves as a shortcut or biased anchor: optimization remains close to a convenient analytical signal even when its calibration has shifted. Direct prediction is more stable when training already contains real-road graphs. Consequently, whether a mathematical prior helps is an empirical distribution-shift question, not an architectural guarantee.

The study also separates the explicit prior from Fiedler node information. Neither Fiedler nor coordinate removal produces a stable loss, whereas replacing the residual architecture does. The robust ingredient in the first protocol is therefore how mathematical and learned estimates are combined, not merely the presence of an eigenvector channel. Targeted-failure correction slopes show that a biased anchor is a measurable calibration failure, while the non-monotone eigengap result warns against using eigengap alone as a trust rule. The poor Deep Sets ranking in several regimes supports adjacency-aware propagation, while the summary MLP shows that coarse graph statistics are insufficient.

6.1 Limitations

Thirteen purposively selected neighbourhoods in six Asian countries improve geographic and morphological coverage but do not establish global or city-scale predictive generality; they span only 48–1,259 nodes. Their size, density, country, and morphology remain partly confounded. The synthetic geometric generator and large-scaling graphs are controlled proxies, not metropolitan OSM accuracy datasets. Spatial clusters are not calibrated to flood, landslide, earthquake, construction, or conflict data. Algebraic connectivity is structural: it omits direction, demand, congestion, capacity, travel time, accessibility, and restoration. Only five seed realizations per area limit within-area precision, while 13 areas limit geographic inference. The hierarchical analysis correctly treats area as the outer unit, but cannot compensate for purposive selection. No probabilistic predictive intervals are produced. Sparse batched neural implementations and spectral sparsification [31] may change runtime rankings at still larger scale.

Future work should pre-register larger metropolitan regions, use nested blocked validation across countries, and factor graph size from density. Hazard rasters could define correlated failures, while traffic assignment could test whether structural loss predicts service degradation. Mixture-of-experts or learned gating may decide when to trust the prior; conformal methods could add graph-population prediction regions. These extensions should preserve area-level replication rather than inflate sample size with correlated scenarios.

7 Conclusion

Residual spectral graph learning offers an interpretable, fast estimator of finite multi-edge connectivity loss, but its advantage depends on training domain. The method improves point estimates across GCN, GraphSAGE, and edge-MPNN in the expanded zero-shot study, with area-level support in three of nine conditions; direct GCN is more stable under blocked OSM-to-OSM transfer. Reporting both outcomes, along with ablations, hierarchical uncertainty, targeted attacks, eigengap and second-order controls, correction calibration, error cases, and separated runtime costs, provides a reproducible account of when a Fiedler prior helps and when it does not.

Data and code availability

Code, cached OSM-derived GraphML, manifests, experiment scripts, environment versions, compact result tables, and release metadata are available as ResiliRoad version 1.0.0 [34]. OSM data are licensed under the Open Data Commons Open Database License and are attributed to OpenStreetMap contributors [33]. Release metadata are included so that the exact submitted version can be archived and cited persistently without relying on a mutable development branch.

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Conflict of interest

The author declares no conflict of interest.

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Generative AI tools were used for language editing and software-development assistance. The author designed the study, verified the implementation and results, and takes responsibility for the manuscript.

A Reproducibility details

All models use AdamW, mean-squared error, validation early stopping, and one PyTorch compute thread. Core runs use at most 80 epochs; blocked OSM transfer uses at most 25. Software versions are Python 3.13.11, NumPy 2.4.4, SciPy 1.17.1, NetworkX 3.6.1, pandas 2.2.3, scikit-learn 1.9.0, PyTorch 2.11.0 (CPU), Matplotlib 3.11.1, OSMnx 2.1.1, and GeoPandas 1.1.4. The machine has 16 logical AMD64 CPUs; CUDA was unavailable. Scripts record exact seeds and emit scenario predictions, training histories, fold assignments, runtime summaries, bootstrap tables, calibration diagnostics, and worst cases. The architecture-control runs use identical data and training settings for GCN and GraphSAGE. Expanded runs add an edge-MPNN whose directed messages receive normalized conductance, inverse-conductance length, failed-edge status, and normalized squared Fiedler difference. Targeted scenarios sample without replacement in proportion to intact-graph edge betweenness. For each OSM area, 20 scenarios per mode and seed are evaluated; hierarchical resampling uses area as the outer cluster and seed within area. Second-order perturbation uses the lowest 12 Laplacian modes, and eigengap tertiles are assigned at graph level. Large-scale runtime graphs use a normalized near-square lattice, four-neighbour grid backbone, independent northwest diagonals with probability 0.12, and inverse-length conductance; the deterministic grid guarantees a connected base graph and yields mean degree approximately 4.2. Sparse `eigsh` uses a fixed starting vector. The intact setup cost is reported both separately and amortized over 1,000 scenarios, preventing preprocessing from being hidden in the speed comparison.

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